

EECS 4080 Computer Science Project

Project Contract

Course: EECS 4080

Academic Year: 2026

Project Title: AI-Generated Image Detection: Evaluating Generalisation Across Generative Model Generations.

Supervisor: Mona Nasery (monan@yorku.ca)

Student: Krishi Rajeshkumar Shah (krishi12@my.yorku.ca)

Student Number: 220968905

Term: Summer 2026

PROJECT DESCRIPTION

AI-generated images have become increasingly difficult to distinguish from authentic photographs. Detection tools have been developed to address this, but they are predominantly trained and evaluated on datasets of images generated by older generative models. As newer architectures such as Stable Diffusion 3, Midjourney v6, and GPT-4o produce increasingly realistic outputs, a critical and largely unanswered question emerges: do existing detectors still work, and if not, why?

This project investigates the generalisation gap in AI-generated image detection — the degree to which a detector trained on one generation of synthetic images degrades when tested against images produced by newer, more capable models. The hypothesis is that current detectors rely on low-level statistical artifacts specific to the architectures they were trained on, and that these artifacts are no longer reliably present in outputs from state-of-the-art generative models. Confirming and quantifying this gap is a meaningful research contribution, as it directly informs where future detection research needs to go.

The project will proceed in two phases. The first phase involves building a strong baseline detector: a fine-tuned EfficientNet-B3 trained on the CIFAKE dataset, augmented with confidence calibration via temperature scaling and Grad-CAM visualisations to make model decisions interpretable. The second phase is the core research contribution: systematically evaluating this detector against images generated by progressively newer models, analysing where and why performance degrades, and identifying which visual features the model relied upon that newer generators no longer produce.

All components will be built using freely available tools and datasets. Training will be conducted on Google Colab's free GPU tier. The final deliverable will include an interactive Gradio demo, a structured evaluation of results across model generations, and a written analysis of findings.

Research Question: How well do AI-generated image detectors trained on current benchmark datasets generalise to images produced by next-generation generative models, and what accounts for the performance gap?

Core components:

- Baseline Detector: Fine-tuned EfficientNet-B3 on CIFAKE (60,000 real and AI-generated images) targeting 90%+ validation accuracy.
- Confidence Calibration: Post-hoc temperature scaling to produce statistically reliable confidence scores, evaluated using Expected Calibration Error (ECE) and reliability diagrams.
- Visual Explainability: Grad-CAM heatmaps identifying which image regions drove classification decisions, used to analyse what the model is and is not detecting across model generations.
- Generalisation Evaluation: Systematic testing of the trained detector on images from progressively newer generative models (CIFAKE baseline, then StyleGAN, Stable Diffusion, Midjourney v6, GPT-4o outputs). Performance drop is measured and attributed.
- Interactive Demo: A Gradio web interface where any image can be submitted and analysed in real time, showing the classification, confidence score, and Grad-CAM heatmap.

BACKGROUND REQUIREMENTS

The student brings substantial practical experience in machine learning, data engineering, and software development, acquired through academic coursework, industry internships, and independent project work. The following summarises the relevant background the student will draw upon throughout this project.

Industry Experience:

- Software Developer Intern, Ontario Government (Jan 2025 – Dec 2025): Developed and maintained Java-based applications, implemented automated test suites, managed CI/CD pipelines on Azure DevOps — directly applicable to building a reproducible, well-tested ML pipeline.
- Machine Learning & Data Science Intern, NFOX Tech (Aug – Nov 2023): Built and evaluated ML models using BERT, scikit-learn, and TensorFlow on real-world text

datasets. Conducted hyperparameter tuning, feature engineering, and model performance evaluation using accuracy, precision, and F1-score metrics.

- Data Engineer Intern, Code Clause (Jan – Jun 2023): Designed Python-based ETL pipelines on Azure Cloud, built SQL database infrastructure, and created Power BI dashboards — background directly applicable to dataset management and evaluation workflows in this project.

Independent Projects:

- Financial Forecasting Model: Built a predictive model achieving an R2 score of 0.87 using XGBoost, pandas, and scikit-learn, with time-series cross-validation and Bayesian hyperparameter optimisation.
- ML Model Bias & Fairness Analysis Platform: Analysed fairness metrics across demographic groups using Fairlearn and scikit-learn; developed interactive visualisations in Tableau and matplotlib.

Technical Skills Relevant to this Project:

- Languages: Python (primary), SQL, Java, JavaScript
- ML Libraries: scikit-learn, TensorFlow, PyTorch, XGBoost, BERT, MLflow
- Data & Cloud: pandas, NumPy, Azure ML Studio, Azure Databricks, AWS, GCP BigQuery
- Tools: Git, Docker, CI/CD pipelines, Jupyter, FastAPI, Agile/Scrum
- Leadership: Lead Developer, York University Data Science Club — led a 12-member team across 5+ ML projects

RESOURCES

Software:

- Python, PyTorch, torchvision, OpenCV, scikit-learn
- pytorch-grad-cam for visual explainability
- scipy for temperature scaling and calibration metrics
- Gradio for the interactive web demo
- Google Colab for GPU-accelerated model training (free tier)

Hardware:

- Personal computer for development and testing
- Google Colab free tier (T4 GPU) — estimated two to three hours of training time

Datasets (all publicly available at no cost):

- CIFAKE — 60,000 labeled real and AI-generated images (Kaggle) — primary training set
- FaceForensics++ — facial manipulation dataset for supplementary testing
- GenImage — images from DALL-E, Midjourney, and Stable Diffusion for generalisation evaluation
- Publicly available image sets from newer models (GPT-4o, Midjourney v6) collected for evaluation purposes

READINGS

- Rossler, A. et al. (2019). FaceForensics++: Learning to Detect Manipulated Facial Images. ICCV 2019.
- Wang, S. et al. (2020). CNN-Generated Images Are Surprisingly Easy to Spot...For Now. CVPR 2020.
- Selvaraju, R. et al. (2017). Grad-CAM: Visual Explanations from Deep Networks via Gradient-based Localization. ICCV 2017.
- Guo, C. et al. (2017). On Calibration of Modern Neural Networks. ICML 2017.
- Bird, J.J. & Lotfi, A. (2024). CIFAKE: Image Classification and Explainable Identification of AI-Generated Synthetic Images. IEEE Access, 12, 15642-15650.
- Corvi, R. et al. (2023). On the Detection of Synthetic Images Generated by Diffusion Models. ICASSP 2023.
- Gagnaniello, D. et al. (2021). Are GAN Generated Images Easy to Detect? A Critical Analysis of the State of the Art. ICME 2021.

DELIVERABLES

- A trained EfficientNet-B3 detector with calibrated confidence scores and Grad-CAM visualisations
- A structured generalisation evaluation report comparing detector performance across four or more generative model generations
- An interactive Gradio demo accepting any uploaded image and returning classification, confidence, and heatmap outputA final report covering the research question, methodology, experimental results, and conclusions
- Documented source code, GitHub repository, and user setup guide

MILESTONE LIST (MAY - AUGUST 2026)

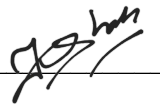
Period	Tasks
May 1 – May 20	• Conduct literature review covering CNN-based detection methods, generative model evolution, explainability (Grad-CAM), and calibration techniques. • Identify research gap. • Set up GitHub repository and Colab environment. • Download and preprocess CIFAKE dataset. Define evaluation metrics (Accuracy, AUC, ECE). Prepare background summary and preliminary project specification.
May 21 – June 15	• Develop baseline detector by fine-tuning EfficientNet-B3 on CIFAKE. • Perform hyperparameter tuning and achieve stable validation performance. • Implement temperature scaling for confidence calibration. • Begin initial evaluation of model performance.
June 16 – July 5	• Integrate Grad-CAM into the model pipeline. • Generate and analyze heatmaps to understand feature importance. Refine calibration using Expected Calibration Error (ECE) and reliability diagrams. • Ensure evaluation pipeline is stable and reproducible.
July 6 – July 20	• Conduct generalisation study by evaluating the detector on images from StyleGAN, Stable Diffusion, Midjourney v6, and GPT-4o. Measure performance degradation and analyze failure cases using Grad-CAM. • Submit Progress Report summarizing achievements, challenges, and any adjustments to project direction.
July 21 – August 10	• Develop and integrate Gradio-based interactive demo. • Combine model, calibration, and explainability components into a unified system. • Perform end-to-end testing and refine user interface.
August 11 – August 28	• Write final report including methodology, experiments, results, and conclusions. • Prepare visualizations and analysis. • Finalize codebase, documentation, and GitHub repository. Prepare presentation and demo. Submit all deliverables.

EVALUATION

Criterion	Weight
Research Quality & Findings — Rigour of the generalisation study, depth of analysis, and significance of conclusions drawn from cross-generation evaluation.	20%
Technical Implementation — Accuracy of the detection model, correctness of calibration, and reliability of Grad-CAM outputs across all test conditions.	20%
Presentation & Demonstration — Quality of live demo, clarity of research narrative, and handling of supervisor questions.	30%
Final Report — Technical depth, experimental rigour, quality of discussion and future directions.	20%
Code Quality & Documentation — Modular, well-commented code with a comprehensive README, setup guide, and consistent GitHub history.	10%

SIGNATURES

Student: Krishni Rajeshkumar Shah: _____



Supervisor: _____



Course Coordinator: _____